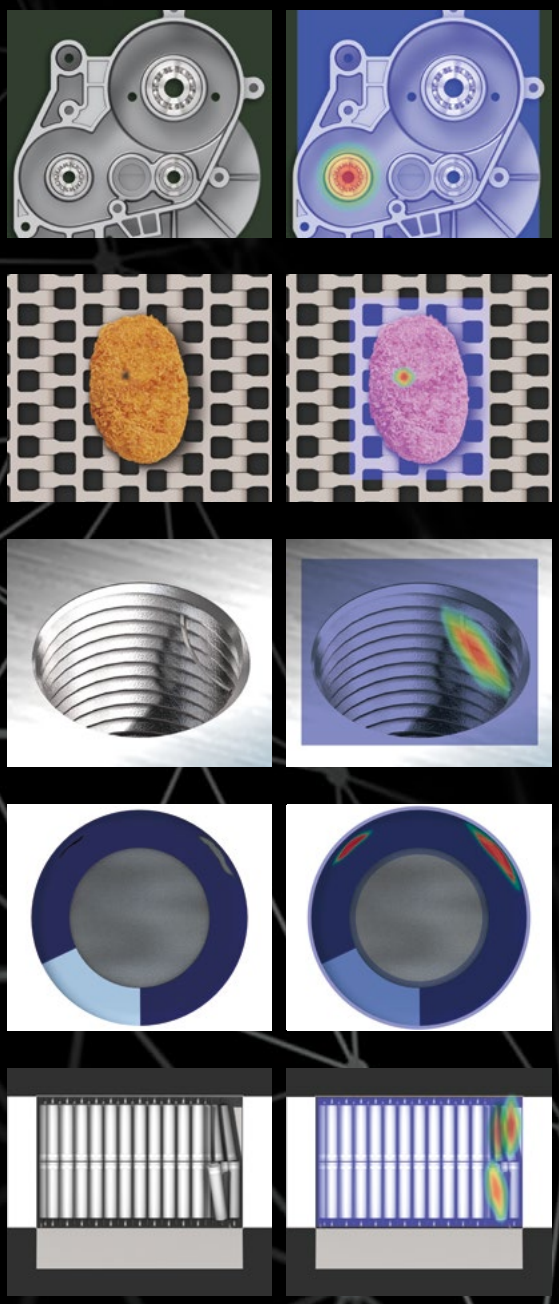
KEYENCE

AI and Vision System

Application Examples



Vision System
with Built-in AI

NEW VS Series

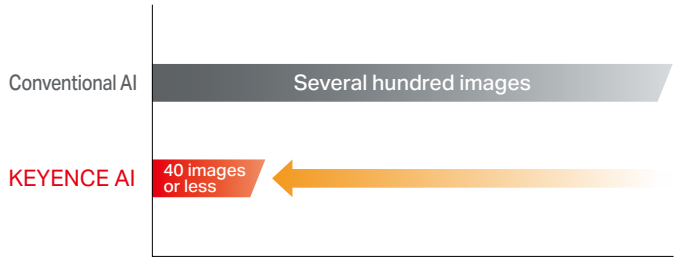
Key points of KEYENCE's AI that can be used in factory automation

1 Unparalleled start-up speed

› Immediate visualization of results

At KEYENCE, we have developed a proprietary learning algorithm by leveraging the on-site expertise accumulated through our direct sales system.
We have developed an AI with maximum performance that learns from a small number of images.

Number of images required for learning before operation



KEYENCE's AI is ready for operation after learning from several to dozens of images

	AI Classification	AI Detection	AI Segmentation	Typical deep learning
Number of images learned	2 to 5 images	40 images or less	40 images or less	100+ images
Learning time	10 seconds	10 seconds to several minutes	10 seconds to several minutes	Several hours to days
GPU	Not required	Not required	Not required	Required

2 Full range of tools to support consistent results without human intervention

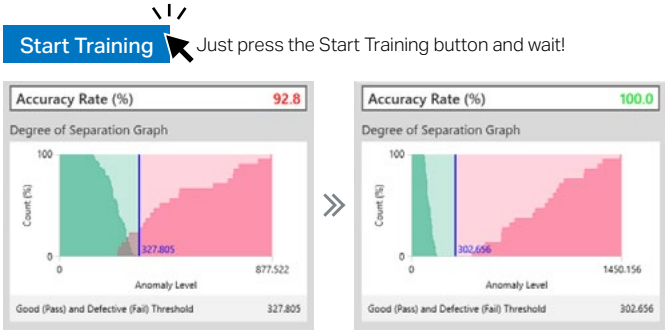
(no experience or skills required)

› No overspecialized tasks

Automatic selection of images for learning

AI Auto Image Selector

The software automatically selects the images needed for learning, so no prior knowledge is required and anyone can easily train the AI.



The Auto Image Selector automatically repeats learning until clear separation of results is possible.

Just paint the defects intuitively

Quick annotation

Simply draw shapes to match the defect shape to train the AI on the defective area.

Paint Paint Polygon

3 Multiple AI tools to choose from depending on detection content × rule-based tools

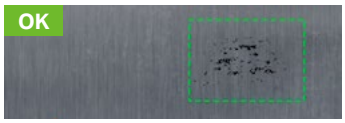
› Flexible combination of optimal tools to best suit your application and purpose

Supervised AI learning detects only targeted defects NEW

What is supervised AI learning?

This algorithm detects defects by separating them from the background, focusing specifically on defects similar to the learned defect data. Since AI Segmentation learns the characteristics of defective areas, it can accurately identify only the targeted defects without excessive false detections. This precise detection capability is highly effective for improving yield rates.

Supervised AI learning
AI Segmentation



Highly accurate detection of defects learned as defective parts.
Other areas of contrast in the image are not mistakenly detected as a defect.

Unsupervised AI learning detects unexpected defects

What is unsupervised AI learning?

This algorithm detects items as defective when they are different from the learned features of good parts. Because it can detect unknown defects, it is effective when it is not possible to prepare all the defective parts and helps prevent the outflow of defective parts.

Unsupervised AI learning
AI Detection

By learning from good parts, the algorithm detects any items that differ from the standard as NG.
It can also identify foreign material contamination, the presence of different product types, and other unexpected issues that may arise after installation as defects.

Easy to differentiate products by simply registering OK and NG products

What is AI Classification?

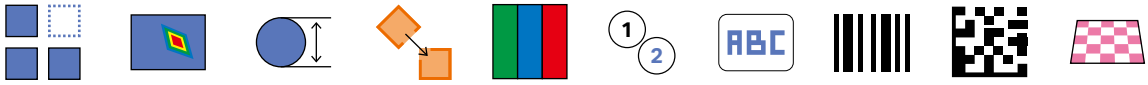
Configuring settings is as simple as registering OK and NG examples. KEYENCE's proprietary AI can then configure the best settings automatically from the registered examples. OK/NG classification and the product type identification inspections conventionally performed with a combination of various tools can now be easily performed with a single tool.

Specialized for differentiation
AI Classification

Simply register an OK product and an NG product

Proven rule-based tools

Robust detection algorithms. Detection capability can be increased by fine-tuning set parameters.



Presence check | Scratches | GD&T measurement | Positioning | Color inspection | Count | Character recognition | 1D code | 2D code | Calibration

Examples of individual AI tool applications and their integration with rule-based inspection tools

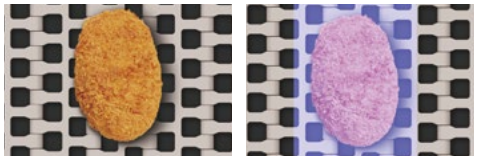
1 Unexpected defects can be detected

Target: frozen food

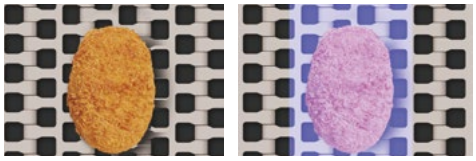
> Using AI Detection (unsupervised AI learning)

With the VS Series, the AI Detection tool can detect unexpected defects. By learning differences in shape and background information, the AI can detect defects such as burned food and damaged products.

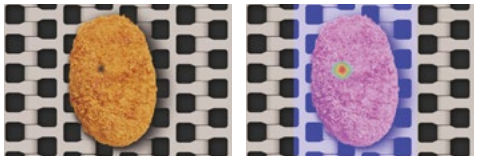
Good product (1)



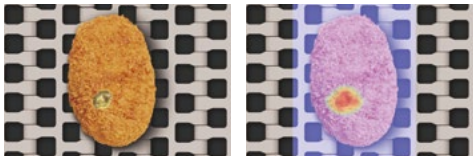
Good product (2): wrong shape



Defective product (1): burned



Defective product (2): missing breading

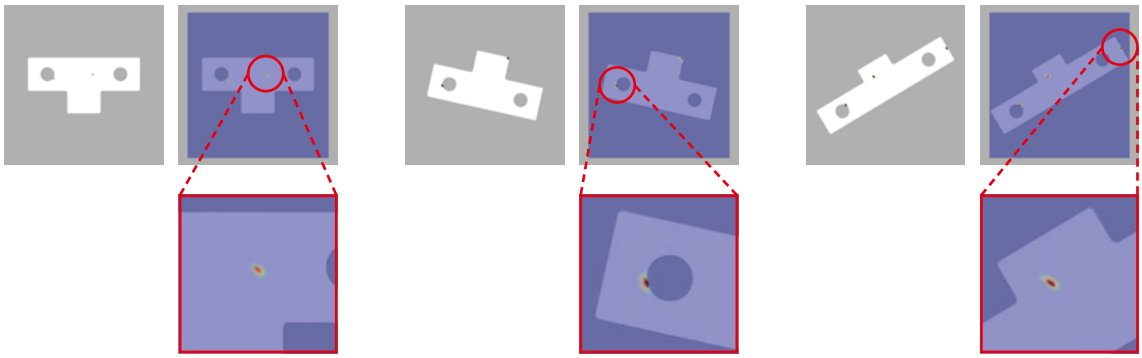


2 Capable of detecting defects in the variation of good parts, which is difficult to isolate with rule-based tools

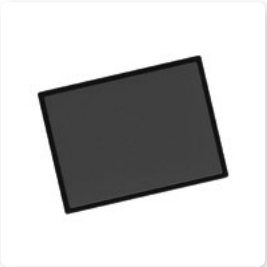
Target: metal parts

> Using AI Segmentation (supervised AI learning)

With the VS Series, the AI Segmentation tool only detects areas similar to learned defects and can do so with high accuracy. Therefore, it is possible to detect only specific dents, ignoring surface contamination and other variations between good parts. Also, since the AI is able to detect specific defects regardless of the shape of the target, a single tool can be used for a wide variety of products.



3 AI and rule-based tools can be used together



Target: chip components

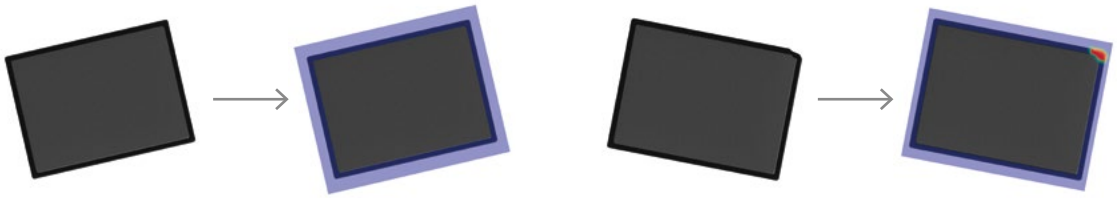
(1) I want to pinpoint the area to be inspected by AI.

> The VS Series AI tools can use the search function of rule-based tools to track the inspection area to the location of the target.

* Position adjustment is effective in improving detection accuracy and reducing processing time.

Set inspection area for AI + set position adjustment with search function of rule-based tools.

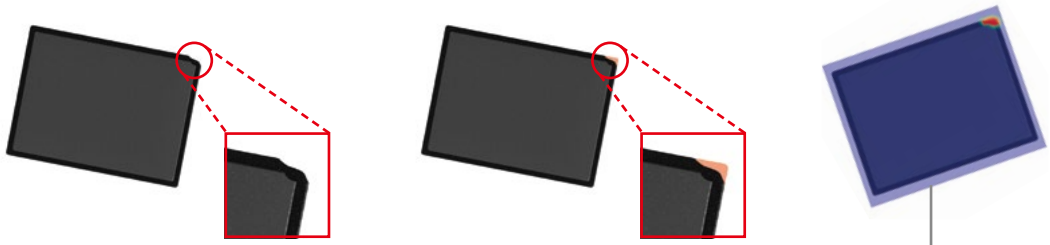
Inspection area tracks the target even if the target is misaligned.



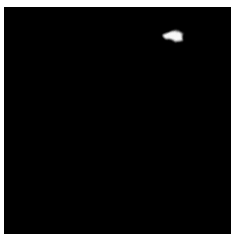
(2) I want to set thresholds for areas detected by AI.

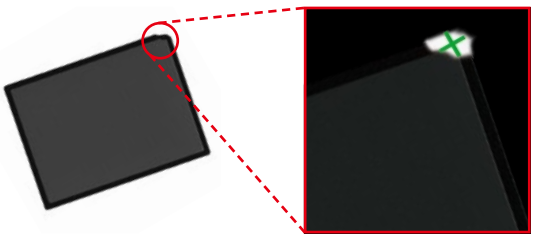
> With the VS Series AI tools, the areas detected by the AI tools can be inspected again with rule-based tools. Subsequent inspection using rule-based tools enables threshold control, maintaining the same approach as conventional methods.

I only want to detect cracks that exceed a certain size. → Annotate defective areas. → Detect the cracks learned with AI Segmentation.



Heat-map image output of areas detected by AI Segmentation. → Dimensional measurement with rule-based tools is possible!





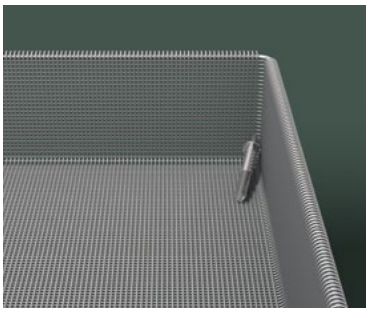
Area:
192 pix
X dimension:
18 pix
Y dimension:
12 pix

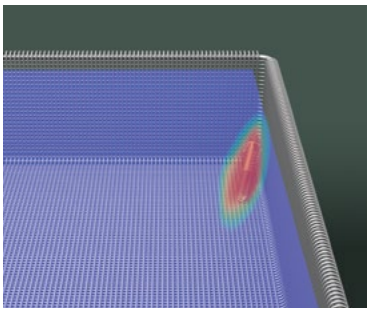
Automotive/Metals

Detection of foreign objects

AI Detection

This inspection is designed to detect remaining product used in processes such as plating. However, the inconsistent pallet shapes and uniform target color make stable detection challenging. With the VS Series, Fine HDR technology captures clear images, enabling AI Detection to reliably identify any remaining cargo.

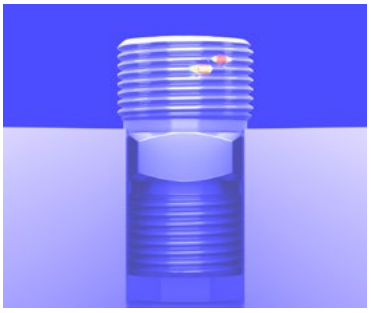




Thread inspection on screws

AI Segmentation

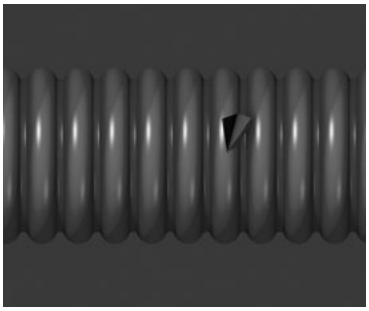
In the inspection for dents on screw threads, it was difficult to detect dents with rule-based tools due to glare and shadows, even when lighting conditions were specially adjusted for this purpose. Training AI Segmentation on dents makes it possible to detect these defects.

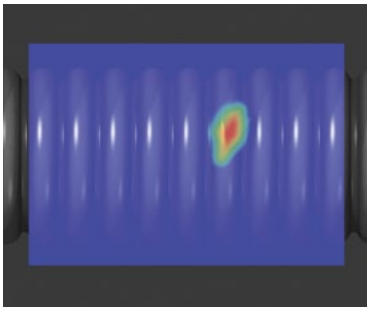


Defect inspection on rubber products

AI Segmentation AI Detection

One of the key properties of rubber is its flexibility to easily take on different shapes. However, this causes challenges with imaging, such as glare and shadows, which can occur despite optimized lighting conditions. AI can effectively detect defects like cracks and blisters, ensuring reliable inspection results.

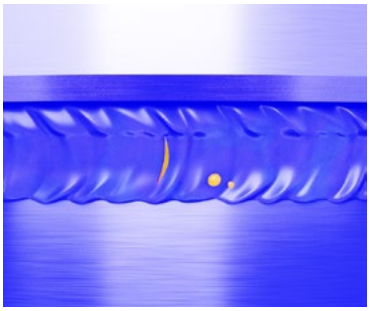




Weld inspections

AI Segmentation AI Detection

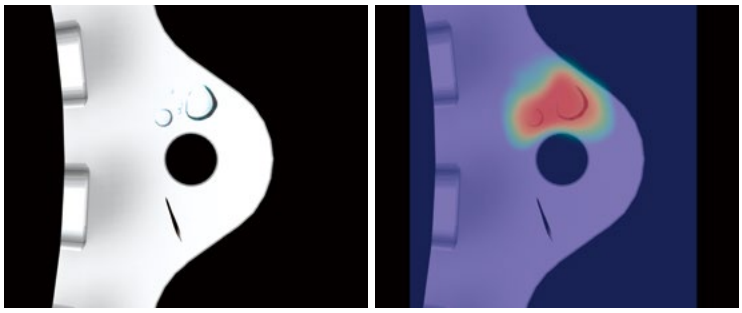
Setting vision system programs was challenging due to the wide variation in welding shapes and colors, as well as the diverse types of defects. By combining AI Detection with AI Segmentation, it becomes possible to achieve highly accurate detection of blowholes exhibiting large variations and spatter traces that have more defined characteristics.



Stator fluid leak inspection

AI Detection

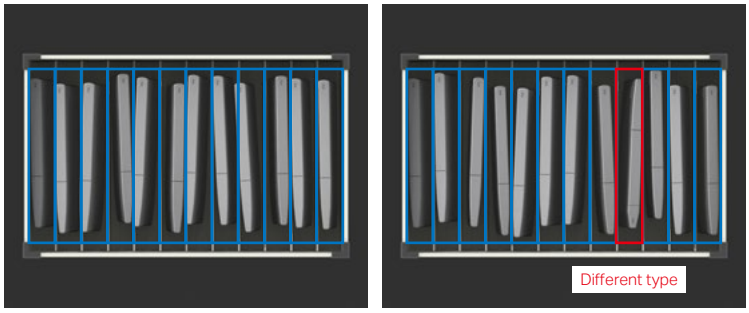
The inspection is affected by changes in color and surface contamination of the stator itself, and there is a limit to the ability to detect only liquid leaks with conventional rule-based tools. AI Detection can take into account this variation and detect only liquid leaks.



Type differentiation inspection on a pallet

AI Classification

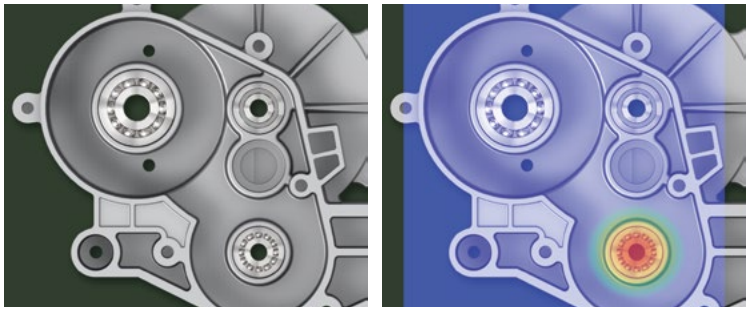
Optimizing settings to differentiate products that appear similar when placed on pallets is challenging due to variations between pallets and changes in product orientation. AI Classification enables stable inspection by learning from just a few images of each product, ensuring reliable differentiation despite these variations.



Transmission bearing product type identification

AI Detection

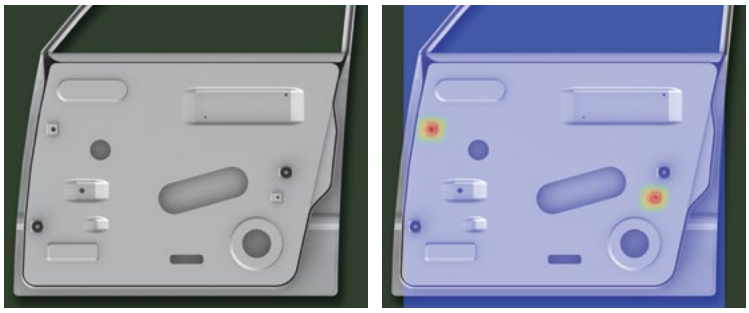
While the inspections appear to be simple, the large volume of parts leads to missed product types during visual inspection. Additionally, the appearance of transmission products changes with each lot, making adjustments difficult with traditional rule-based tools. However, by using AI Detection, inspections can be easily performed within a single area, improving accuracy and efficiency.



Door panel assembly confirmation

AI Detection

With conventional rule-based tools, inspections for each door panel required defining detailed inspection areas specific to each panel type. However, by utilizing AI Detection, various assembly confirmations can be performed within a single inspection area. For more detailed inspections, ZoomTrax enables switching between different fields of view, allowing flexible and thorough detection.

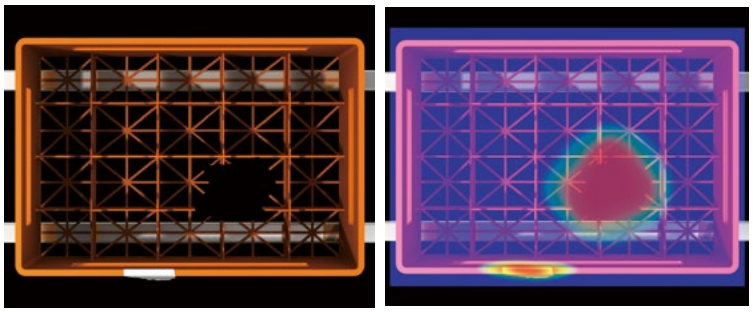


Food / Pharmaceuticals / Commodities

Appearance inspection of crates, cases, and totes

AI Detection

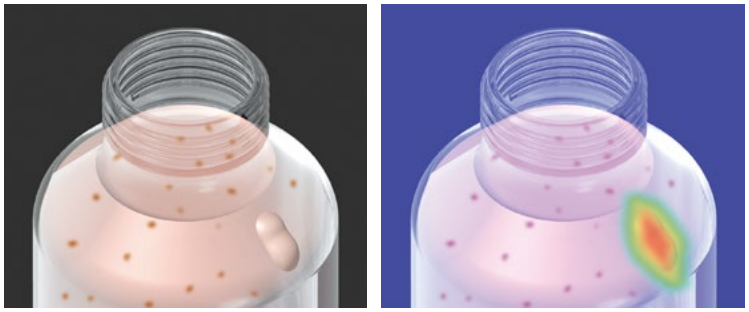
Inspecting seal residue after cleaning was challenging with conventional rule-based tools due to inconsistent product shapes. AI Detection can reliably detect seal residue, deformation, cracks, and similar defects with stable accuracy.



Inspection for spillage

AI Detection

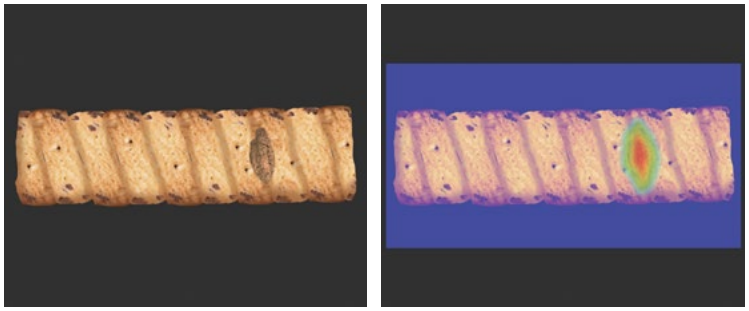
Detecting spills and leaks on packaged goods can be challenging due to low-contrast differences between the package and spilled material. AI Detection identifies defects with low contrast differences while learning variations between non-defective targets.



Food product inspection

AI Detection

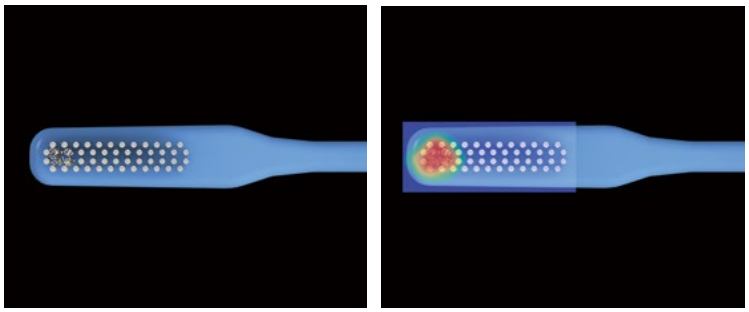
This is an example of an inspection that is difficult to perform with rule-based tools due to variations in color, shape, and pattern. With the VS Series, the defective parts of the product are first emphasized in imaging, then combined with AI Detection to achieve stable inspection.



Toothbrush bristle tip appearance inspection

AI Detection

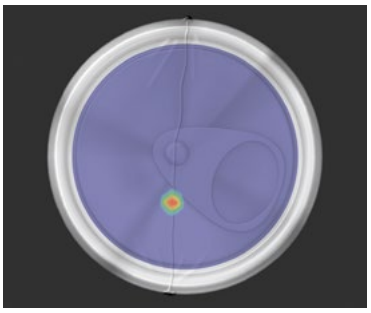
Due to the variation in the shape of toothbrush bristle tips, inspections were difficult to perform using conventional rule-based tools. With AI Detection, stable inspections are possible as the AI tool identifies differences from good parts.



Inspection for shrink wrap defects

AI Detection

Wrinkles and glare make it difficult to detect defects in transparent or patterned shrink wrap. AI Detection not only allows for stable inspection, but also displays a heat map to view the defective areas.



Wrinkle detection on labels and packaged products

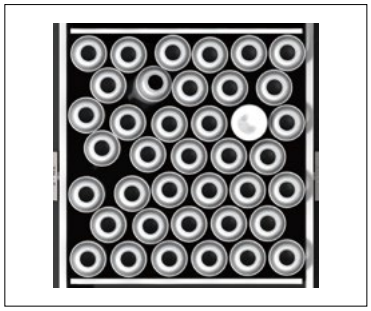
AI Segmentation

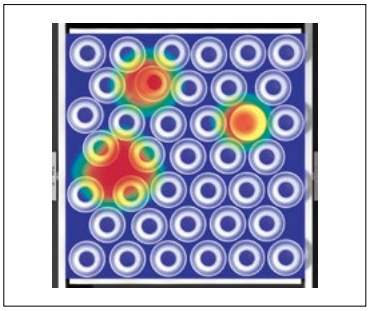
Detecting wrinkles on packaged consumer goods is an example of a traditionally difficult inspection. KEYENCE's proprietary LumiTrax imaging technology emphasizes the wrinkled areas, enabling stable detection with AI Segmentation.

Packaging process inspection (alignment inspection)

AI Detection

Alignment inspection is an example of an inspection that appears simple but can be quite difficult. AI Detection is optimal for this inspection due to the large number of unexpected defects. Defects such as missing, tilted, and reverse-oriented products can be detected in a stable manner.

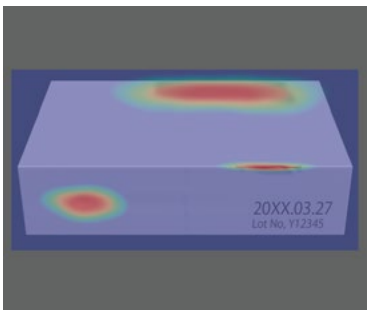




Tear and dent inspection of boxes

AI Segmentation AI Detection

Inspections for tears, dents, and peeling in boxes were one example of inspections that were difficult to perform with rule-based tools, as the pattern of the target itself also affected the inspections. AI Detection can inspect for large dents, while small tears can be trained to be detected with AI Segmentation.

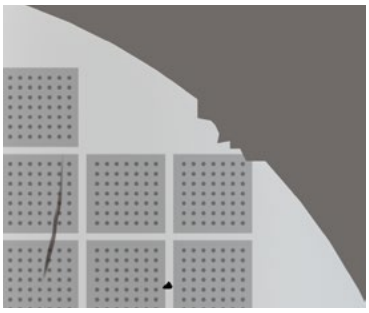


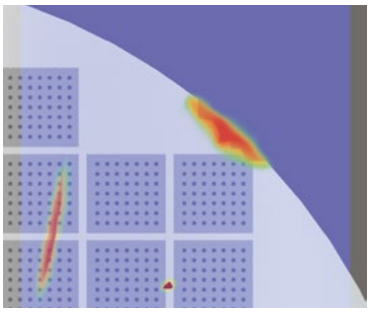
Electric and Electronic Parts

Wafer appearance inspection

AI Segmentation AI Detection

Cracks and chipping on wafers were difficult to detect using rule-based tools. With AI, cracks can be inspected by training AI Segmentation on the specific shapes of defects, while chipping can be detected by identifying differences from good parts with AI Detection.

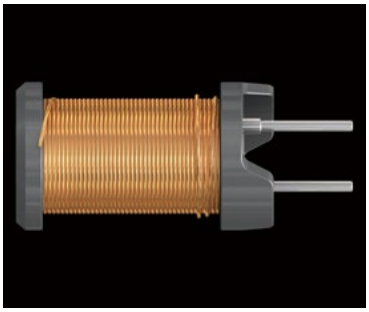


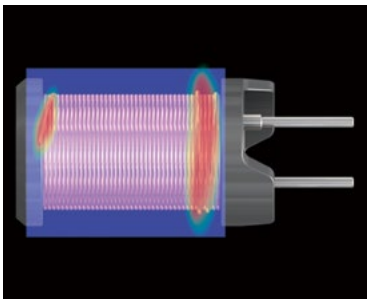


Inductor winding inspection

AI Segmentation AI Detection

Inspection of winding has conventionally been performed using a combination of several algorithms, including edge width and color. With AI Detection, a single tool can detect a variety of winding defects. In addition, for whisker inspections, which require greater accuracy, AI Segmentation can be used to obtain stable detection.

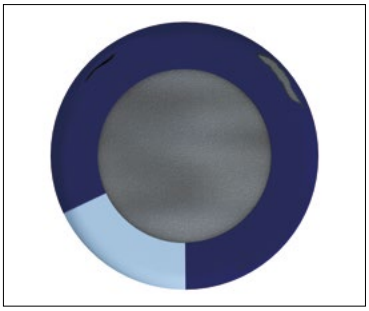


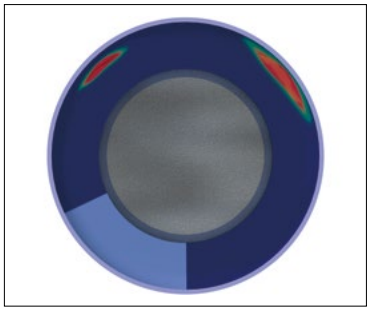


Capacitor sleeve inspection

AI Segmentation AI Detection

Large tears can be detected with conventional rule-based tools, but tears with low contrast are difficult to detect. With AI Segmentation, even small tears can be learned and detected stably.

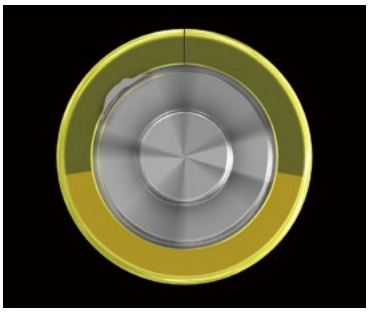


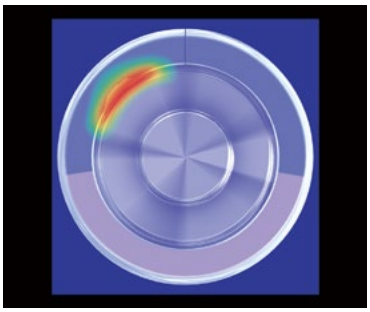


Gasket float inspection

AI Detection

In conventional inspections with rule-based tools, gasket floaters are obscured by irregularities on the surface, making it difficult to adjust the area and resulting in frequent excessive detection. AI Detection enables inspection in a single area, reducing excessive detection and at the same time significantly reducing setup time.

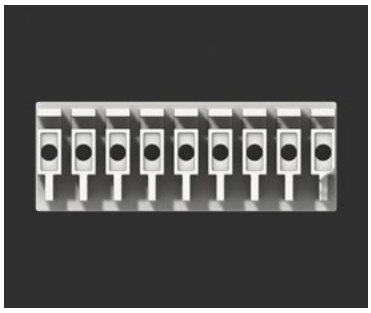


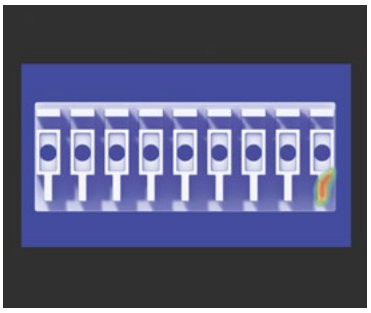


Connector short shot inspection

AI Detection

Although a connector short shot is a defective target, its color remains unchanged, resulting in low contrast. Additionally, connectors often have complex shapes, requiring multiple inspection areas with conventional methods. AI Detection can identify multiple defect modes within a single inspection area. Another major advantage is the ability to combine AI Detection with rule-based inspections such as dimensional measurement and OCR, leveraging the strengths of the VS.

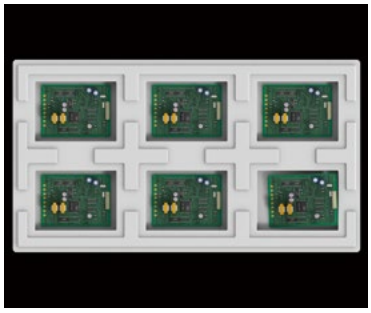


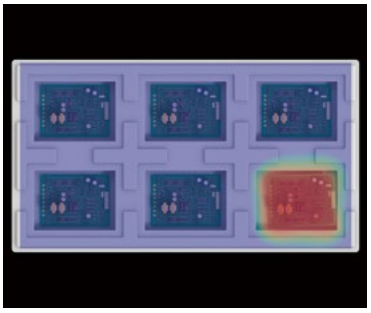


Module alignment confirmation

AI Detection

AI Detection can identify not only module misalignment but also incorrectly stacked trays—defects that are difficult to detect with conventional rule-based tools. While this is often checked visually, the VS Series can automate this process, significantly reducing labor requirements.





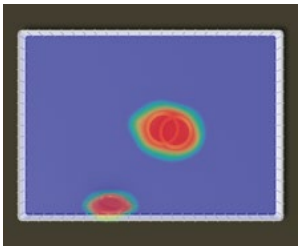
Vision-Guided Robotics × AI

AI is mainly used for appearance inspections and presence inspections, but can also be used with vision-guided robotics.

Detection of overlapped and leftover product

AI Detection

AI Detection can easily detect overlapping or tilted targets, which are difficult to detect with conventional rule-based tool. Alerts can be issued when targets remain but cannot be picked, preventing these remaining targets from moving along to subsequent processes.



Vision guidance + product type identification

AI Classification

AI can simultaneously inspect multiple types of targets while picking, eliminating concerns even in mixed production lines with various targets. This capability is particularly effective for irregularly shaped targets that exhibit significant variations among good parts.

Achieve the inspection you envisioned.

Supervised AI learning × unsupervised AI learning × rule-based tools

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